

SCORING GUIDELINES FOR BONE AGE ESTIMATION

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(Hartnett, 2010)	
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GUIDELINES FOR PUBIC SYMPHYSIS PHASES:

PHASE 1: A clear ridge and furrow system extends from the pubic tubercle onto the inferior ramus. Ridges and furrows are deep and well-defined and do not look worn down. There is no dorsal lipping. Bone is of excellent quality and is firm, heavy, dense, and smooth on the ventral and dorsal body. There is no rim formation. The dorsal plateau is not formed. The ridges and furrows extend to the dorsal edge.

PHASE 2: The rim is in the process of forming, but mainly consists of flattening of the ridges on the dorsal aspect of the face and ossific nodules present along the ventral border. Ridges and furrows are still present. The ridges and furrows may appear worn down or flattened, especially on the dorsal aspect of the face. The furrows are becoming shallow. The upper and lower rim edges are not formed. There is no dorsal lipping. The bone quality is very good, and the bone is firm, heavy, dense, and smooth on the ventral and dorsal body, with little porosity. The pubic tubercle may appear separate from the face.

PHASE 3: The lower rim is complete on the dorsal side of the face, and is complete until it ends approximately halfway up the ventral face leaving a medium to fairly large gap between the lower and upper extremities on the ventral face. This enlarged 'V' is longer on the dorsal side than the ventral side. Some ridges and shallow furrows are still visible, but appear worn down. In some cases, the face is becoming slightly porous. The rim is forming both on the dorsal aspect of the face and the upper and lower extremities. In some cases, there is a rounded buildup of bone in the gap between the upper and lower extremities above the enlarged 'V'. Bone quality is good; the bone is firm, heavy, dense, and has little porosity. The dorsal surface of the body is smooth, and there are small bony projections near the medial aspect of the obturator foramen. The ventral aspect of the body is not elaborate. Very slight to no dorsal lipping. Quality of bone and rim completion are important deciding factors. **Variant:** In some cases, a deep line or epiphysis is visible on the ventral aspect parallel to and adjacent to the face (males only).

PHASE 4: In most cases, the rim is complete at this stage, but may have small ventral hiatus on the superior and ventral aspect of the rim. The face is flattened and not depressed. Remnants of ridges and furrows may be visible on the face, especially on the lower half. The quality of the bone is good, but the face is beginning to appear more porous. The dorsal and ventral surfaces of the body are roughened and becoming coarse. There is a slight dorsal lipping. In females with parturition pits, dorsal lipping can be more pronounced. The ventral arc may be large and elaborate in females.

PHASE 5: The face is becoming more porous and is depressed, but maintains an oval shape. The face is not irregularly-shaped or erratic. The rim is complete at this stage. In general, the rim is not irregular. Ridges and furrows are absent on the face. There may be some breakdown of the rim on the ventral border, which appears as irregular bone (not rounded/solid). The ventral surface of the body is roughened and irregular, with some bony excrescences. The dorsal surface of the body is coarse and irregular. Projections are present on the medial aspect of the obturator foramen. Bone quality is good to fair; it is losing density and is not smooth. The bone is moderately light in weight. In females, the ventral arc is prominent.

PHASE 6: The face is losing its oval shape and is becoming irregular. The rim is complete, but breaking down, especially on the ventral border. The rim and face are irregular, porous, and macroporous. Bone quality is fair, and the bone is lighter and more porous, even with bony buildup on the ventral body surface. The rim is eroding. The dorsal surface of the bone is rough and coarse. There are no ridges and furrows. Dorsal lipping is present. Projections are present at the medial aspect of the obturator foramen. Bone weight is a major deciding factor between phases 6 and 7.

PHASE 7: The face and rim are very irregular in shape and are losing integrity. The rim is complete but is eroding and breaking down, especially on the ventral border. There are no ridges and furrows. The face is porous and macroporous. Dorsal lipping is pronounced. Bone quality is poor, and the bone is very light and brittle. Bone weight is an important deciding factor. The dorsal surface of the bone is roughened. The ventral surface of the body is roughened and elaborate. Projections are present at the medial wall of the obturator foramen. The pubic tubercle is elaborate and proliferative. Bone weight is a major deciding factor between phases 6 and 7.

VARIANT: The rim is complete except for a lytic/sclerotic appearing hiatus at the superior ventral margin that extends toward the pubic tubercle and sometimes underneath the ventral rim, which should not be confused with a hiatus.

GUIDELINES FOR STERNAL END OF THE FOURTH RIB PHASES:

PHASE 1: The pit is shallow and flat, and there are billows in the pit. The pit is shallow U-shaped in cross-section. The bone is very firm and solid smooth to the touch, dense, and of good quality. The walls of the rim are thick. The rim may show the beginnings of scalloping.

PHASE 2: There is an indentation to the pit. The pit is V-shaped in cross-section, and the rim is well defined with round edges. The rim is regular with some scalloping. The bone is firm and solid, smooth to the touch, dense, and of good quality. There is no flare to the rim edges; they are parallel to each other. The pit is still smooth inside, with little to no porosity. In females, the central arc, which manifests on the anterior and posterior walls as a semicircular curve, is visible.

PHASE 3: The pit is V-shaped, and there is a slight flare to the rim edges. The rim edges are becoming undulating and slightly irregular, and there may be remnants of scallops, but they look worn down. There are no bony projections from the rim. There is porosity inside the pit. The bone quality is good; it is firm, solid, and smooth to the touch. The rim edges are rounded, but sharp. In many females, there is a build-up of bony plaque, either in the bottom of the pit or lining the interior of the pit, creating the appearance of a two-layer rim. An irregular central arc may be apparent.

PHASE 4: The pit is deep and U-shaped. The edges of the pit flare outwards, expanding the oval area inside the pit. The rim edges are not undulating or scalloped but are irregular. There are no long bony projections from the rim, and the rim edges are thin, but firm. The bone quality is good, but does not feel dense or heavy. There is porosity inside the pit. In some males, two distinct depressions are visible in the pit. In females, the central arc may be present and irregular, however, the superior and inferior edges of the rim have developed, decreasing the prominence of the central arc.

PHASE 5: There are frequently small bony projections along the rim edges, especially at the superior and inferior edges of the rim. The pit is deep and U-shaped. The rim edges are irregular, flared, sharp, and thin. There is porosity inside the pit. The bone quality is fair; the bone is coarse to the touch and feels lighter than it looks.

PHASE 6: The bone quality is fair to poor, light in weight, and the surfaces of the bone feel coarse and brittle. There are bony projections along the rim edges, especially at the superior and inferior edges, some of which may be over 1 cm long. The pit is deep and U-shaped. The rim is very irregular, thin, and fragile. There is porosity inside the pit. In some cases, there may be small bony extrusions inside the pit. In females, the central arc is not prominent.

PHASE 7: The bone is very poor quality, and in many cases, translucent. The bone is very light, sometimes feeling like paper, and feels coarse and brittle to the touch. The pit is deep and U-shaped. There may be long bony growths inside the pit. The rim is very irregular with long bony projections. In some cases, much of the cartilage has ossified and window formation occurs. In some females, much of the cartilage in the interior of the pit has ossified into a bony projection extending more than 1 cm in length.

VARIANT: In some males, the cartilage has been completely or almost completely ossified. The ossification tends to be a solid extension of bone rather than a thin projection. All of the bone is of very good quality including the ossification. It is dense, heavy, and smooth. In these instances, bone quality should be the determining factor. There are probably other factors, such as disease, trauma, or substance abuse that caused premature ossification of the cartilage. When the individual is truly very old, the bone quality will be very poor. Be aware of these instances where a rib end may appear very old because of ossification of the cartilage but is really actually a young individual, which can be ascertained by bone quality. In these cases, consult other age indicators in conjunction with the rib end.

GUIDELINES FOR AURICULAR SURFACE PHASES:

Phase 1: Billowing with possible striae; mostly fine granularity with some coarse granularity possible.

Phase 2: Striae; coarse granularity with residual fine granularity; retroauricular activity may be present.

Phase 3: Decreased striae with transverse organization; coarse granularity; retroauricular activity present beginnings of apical change.

Phase 4: Remnants of transverse organization; Coarse granularity becoming replaced by densification; retroauricular activity present; apical change; macroporosity is present

Phase 5: Surface becomes irregular; surface texture is largely dense; moderate retroauricular activity; moderate apical change; macroporosity

Phase 6: Irregular surface; densification accompanied by subchondral destruction; severe retroauricular activity; severe apical change; macroporosity

GUIDELINES FOR THIRD MOLAR PHASES:

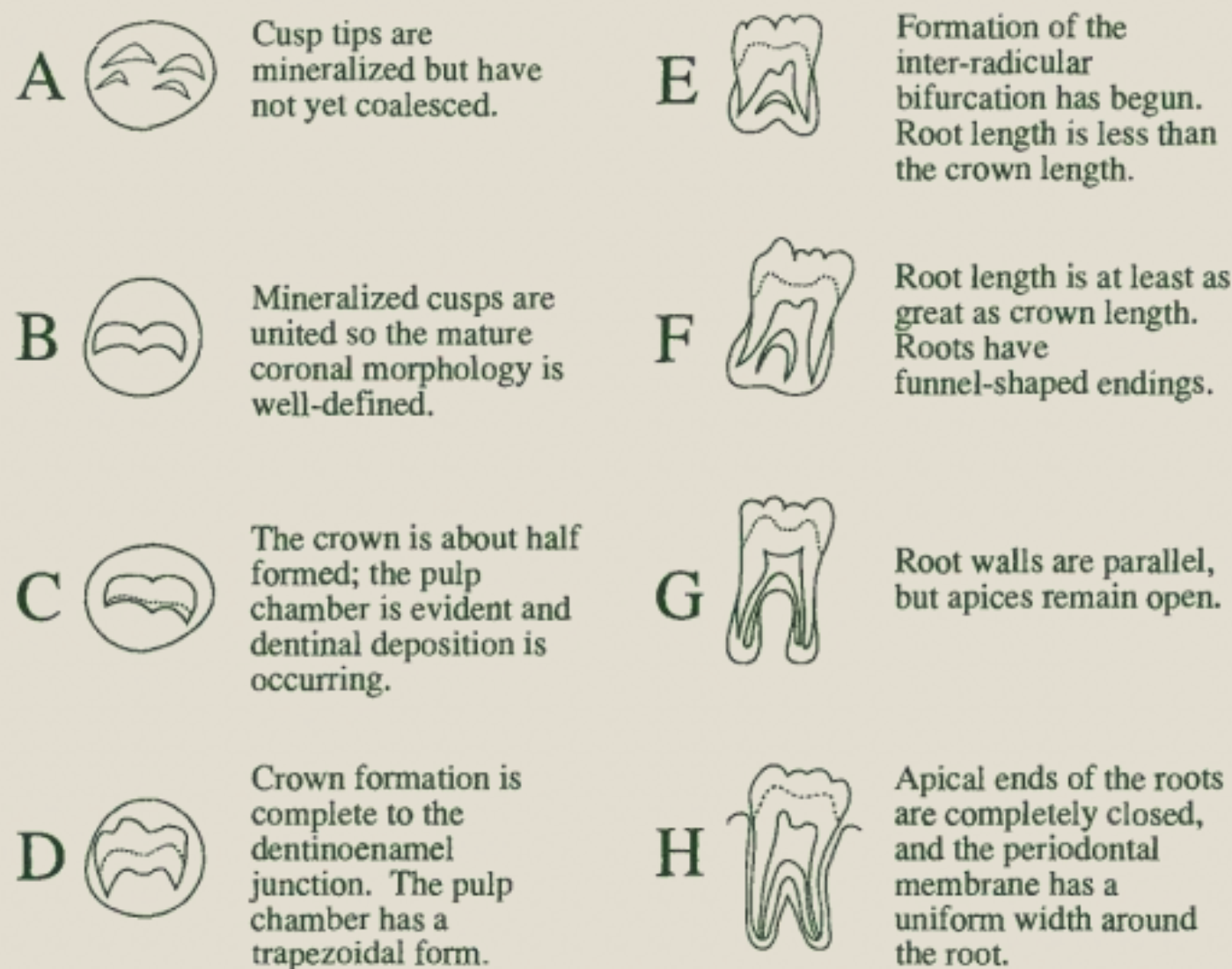


FIG. 1—Schematic drawings and definitions of the eight stages of crown and root formation used to score third molar development (modified from Demirjian et al. [8]). Grades A and B did not occur in the age interval examined (14.1 to 24.9 yrs), and grade C occurred in less than 1% of the sample and was omitted from analysis.

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